

SUPPLEMENTARY MATERIAL: ACOUSTIC OVERSPECIFICATION IN ELECTRONIC DANCE MUSIC TAXONOMY

Weilun XU^{†1}, Tianhao DAI¹, Oscar GOUDET¹, and Xiaoxuan WANG²

¹*School of Computer and Communication Sciences (IC), École Polytechnique Fédérale de Lausanne, Switzerland*

²*Digital and Cognitive Musicology Lab, École Polytechnique Fédérale de Lausanne, Switzerland*

A. FEATURE TAXONOMY

Table 1 maps each feature category to its musicological significance and EDM production relevance.

B. CLUSTER VALIDATION CURVES

Figure 1 shows the four validity criteria across $k=10\text{--}30$: silhouette plateaus across $k=16\text{--}18$ (visual knee around 17), while Davies-Bouldin reaches a local minimum near $k=20$ —its elbow-detected vote ($k=29$) reflects diminishing returns rather than the raw minimum. The broad consensus zone between 17 and 20 confirms that further splitting yields negligible improvement.

C. DJ TOOLS SENSITIVITY ANALYSIS

To verify that DJ Tools does not distort our findings, we re-ran the natural- k consensus after excluding all 100 DJ Tools tracks. The four criteria voted for $k \in \{17, 20, 22, 29\}$ (four-way tie, resolved to $k=17$ by parsimony), compared with the original $k \in \{17, 18, 20, 29\}$. Figure 2 shows that the metric curves nearly overlap with and without DJ Tools, confirming negligible influence. The apparent shift to $k=17$ reflects tie-breaking mechanics across near-identical curves rather than a genuine structural difference.

D. ACOUSTICALLY CONVERGENT GENRE PAIRS

To identify genre pairs that are acoustically near-indistinguishable despite carrying distinct commercial labels, we apply a dual filter: (1) high co-clustering rate under K-means ($k=35$, 50 restarts), requiring that tracks from the two genres frequently land in the same cluster; and (2) a profile difference $\Delta < 15$, computed as the mean absolute difference across the six radar dimensions in Figure 4 of the main text (on the 0–100 scale, $\Delta < 15$ indicates less than 15% average divergence per dimension—a deliberately conservative threshold to retain only genuinely near-identical pairs). Of 595 possible genre pairs, 273 pass

the Δ threshold; applying the co-clustering filter yields the top convergent pairs shown in Figure 3.

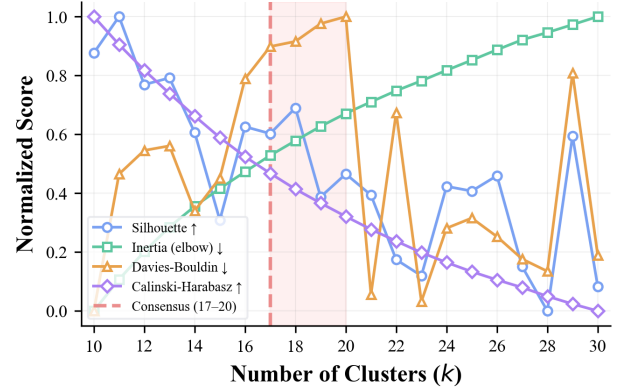


Figure 1: Natural cluster discovery across $k \in [10, 30]$. The vertical dashed line marks the visual consensus elbow around $k=17$, representative of the 17–20 cluster plateau where validation metrics stabilize.

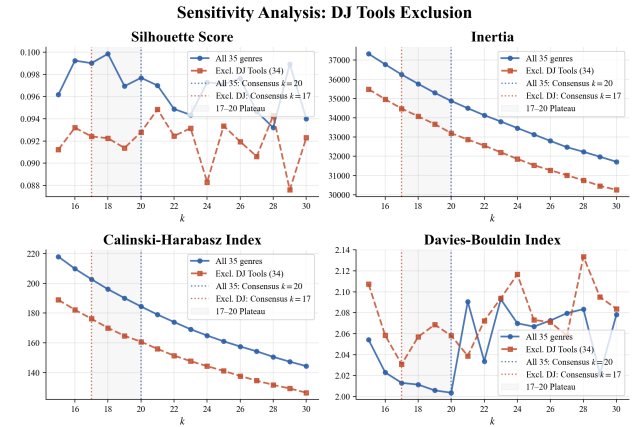


Figure 2: Sensitivity analysis isolating the structural impact of the “DJ Tools” utility category. Global metric curves fundamentally overlap whether DJ Tools are included (blue) or excluded (orange), confirming that this functional category does not distort the underlying optimal- k consensus.

Table 1: Taxonomy of the 194 acoustic features. This mapping bridges low-level signal processing (Feature) with theoretical concepts (Musicological Significance) and practical mixing conventions (EDM Production Relevance), demonstrating how cultural genre rules manifest acoustically.

Category	Feature	Musicological Significance	EDM Production Relevance
Conventional MIR			
Timbral	Spectral centroid, spread, entropy, flux, rolloff; MFCCs; ZCR; energy & energy entropy; spectral flatness & bandwidth	Captures Smalley’s spectral space—brightness, density, and the dynamic evolution of synthesized sounds.	Distinguishes filter automation in Deep House from distortion profiles in Dubstep; identifies synthesis methods (analog vs. FM).
Rhythmic	Tempo/BPM & histograms; autocorrelation-derived danceability; frequency-band beat emphasis; onset rate	Tempo as a primary cultural signifier—from House’s 120–130BPM to D&B’s 160–180BPM; quantifies groove mechanics.	Broad genre boundaries dictated by BPM limits; beat-band profiles reveal sub-bass drops (Trap / Bass) vs. mid-range punch (Techno).
Harmonic	Chroma vectors; chroma deviation	Maps tonal vocabularies and harmonic stasis—essential for EDM’s modal and repetitive practices.	Distinguishes modal, drone-based Techno from jazz-influenced Liquid D&B; measures repetition vs. functional harmonic progression.
Production and Texture			
Dynamics & Texture	H/P ratio, crest factor, sub-bass ratio, sidechain proxy, spectral contrast, RMS statistics, dynamic range, energy variance, spectral modulation, kick prominence	Quantifies the mastering footprint, timbral sharpness, and large-scale structural energy (e.g., 8-bar variance).	Captures brickwall compression (the loudness war), sub-woofer focus, sidechain ducking aesthetics, and structural drops.
Tempogram-based			
Temporal	Fourier, autocorrelation, and cyclic tempograms	Rhythmic spectrogram capturing localized tempo evolution and interlocking periodicities over time.	Reveals build-up acceleration and breakdown deceleration; identifies half-time/double-time rhythmic switches (Dubstep vs. D&B).

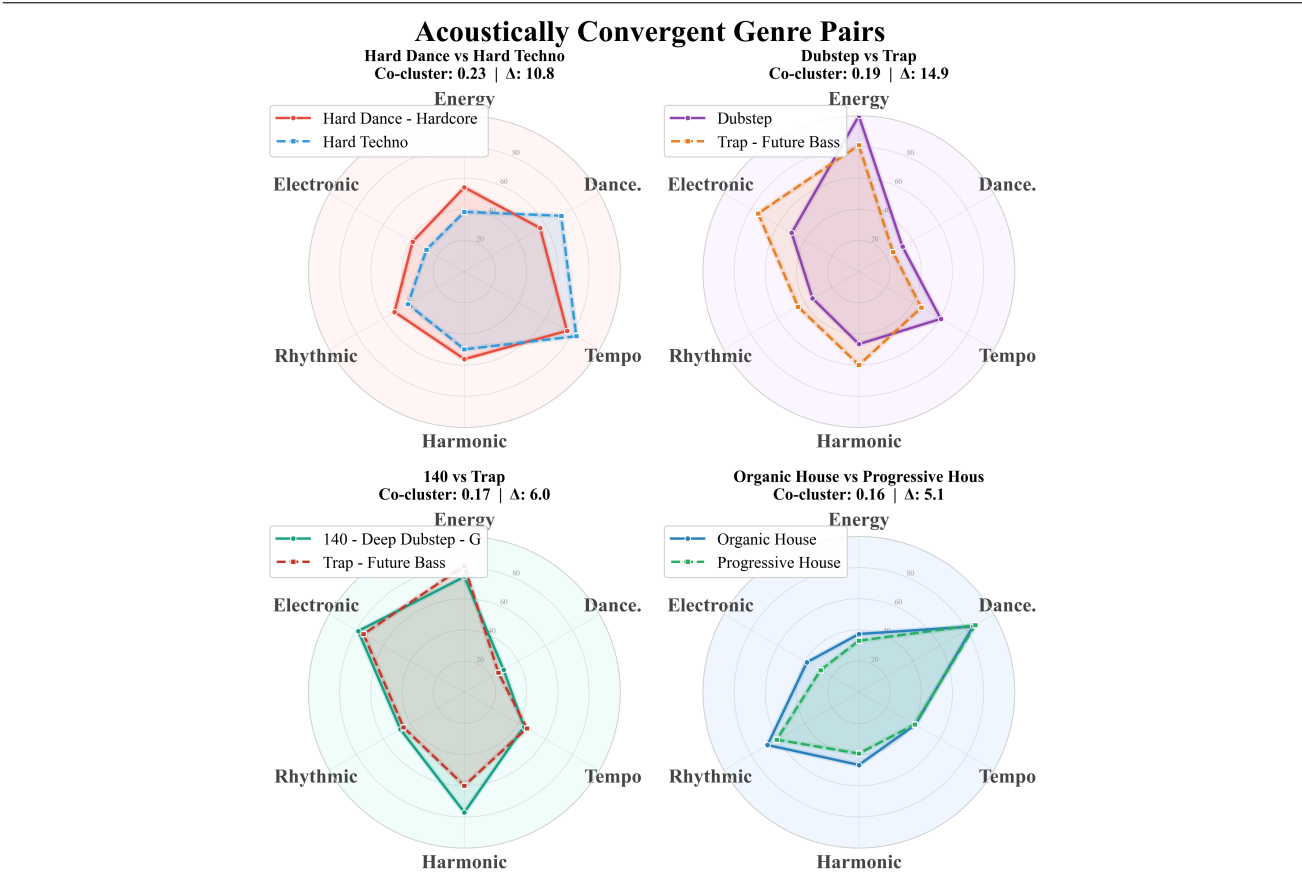


Figure 3: Acoustically convergent “phantom” boundaries identified via dual filtering (high co-clustering frequency and mean profile difference $\Delta < 15$). Each subplot overlays the radar profiles of two commercially distinct genres possessing nearly identical acoustic signatures, illustrating explicit cases where commercial taxonomic boundaries lack concrete acoustic justification.